Candidate Local GR Completion from the Existing Complex-Time Profile

Free-embedding lift, pure-Θ Gauss action, and exact vacuum closure

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Abstract

The torsion-free single-section construction $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta \in W_L$ obeys a concurrent-vector no-go and cannot generate a generic non-flat vacuum geometry. This note proves a precise alternative using no new fundamental field: retain the already postulated periodic complex-time profile $\psi \mapsto \Theta(x,\psi)$ and read the four-dimensional metric from the normalized Haar average of the central Lorentz product. A local free isometric embedding of any Lorentzian four-metric into $\mathbb{E}^{13,1}$ lifts isometrically to fourteen orthonormal W_L-valued profile modes of the same Θ. The ten normal second-derivative profiles are then independent, so variation of the pure-Θ Gauss action is exactly equivalent to the vacuum Einstein–Λ equations on that branch. This closes local representability and local vacuum closure for the profile metric. It does not derive κ, Λ, profile selection, or the profile readout from the previous canonical master action; promotion therefore requires a deliberate canonical decision rather than a silent status change.

Canonical impact

The construction below is mathematically compatible with one fundamental field $\Theta(q,\tau)$ and adds no independent connection or tetrad. It is nevertheless *not compatible with the current wording of AXIOM C*, which forbids compact-ψ averaging as the canonical metric readout. The result is therefore a candidate completion until the repository maintainer either promotes it by a major axiom revision or rejects the profile readout.

1 Central profile pairing

Let

$$W_L = \{ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R}\} \subset \mathbb{B}$$

be the canonical Lorentz slice. For $U, V : S^1_{R_\psi} \to W_L$ define the real symmetric profile pairing by the central identity

$$\langle U, V\rangle_\psi \, \mathbf{1} := \frac{1}{L_\psi}\int_0^{L_\psi} \frac{1}{2}\left(U^\sharp V + V^\sharp U\right) \mathrm{d}\psi, \qquad L_\psi = 2\pi R_\psi. \tag{1}$$

It is the orthogonal direct sum of countably many copies of the Lorentz form on W_L and therefore has infinitely many positive and negative directions.

Definition 1.1 (Candidate profile metric)**.** *For a smooth periodic field $\Theta : U \times S^1_{R_\psi} \to W_L$ set*

$$E_\mu := \partial_\mu\Theta, \qquad \boxed{g_{\mu\nu}[\Theta] = \frac{1}{\mathcal{N}_0}\langle E_\mu, E_\nu\rangle_\psi\,.} \tag{2}$$

Equivalently, $g_{\mu\nu}\mathbf{1}$ is the normalized Haar average of the pointwise central anticommutator. No trace or real-part projection is used.

Proposition 1.2 (Finite Lorentz ambient block inside one profile)**.** *There exist fourteen smooth periodic profiles $F_A : S^1 \to W_L$, $A = 0, \dots, 13$, satisfying*

$$\langle F_A, F_B \rangle_\psi = \eta_{AB}, \qquad \eta_{AB} = \mathrm{diag}(-1, +1, \dots, +1). \tag{3}$$

Proof. Take a unit timelike constant profile for F_0. For the positive directions use normalized sine/cosine profiles in the three spacelike quaternion basis directions, with distinct pairs (Fourier number, algebra direction). Fourier orthogonality makes different choices orthogonal, while the Lorentz slice identity fixes every norm to $+1$. There are arbitrarily many such positive profiles, so thirteen can be selected. □

Let $\iota : \mathbb{R}^{13,1} \to \mathscr{V}_\psi$ denote the isometric linear map

$$\iota(v) := v^A F_A, \qquad \langle \iota(v), \iota(w) \rangle_\psi = \eta_{AB} v^A w^B.$$

2 Free-embedding lift

Definition 2.1 (Free isometric embedding)**.** *A local isometric embedding $Y : U \to \mathbb{E}^{13,1}$ is free when the fourteen ambient vectors*

$$\{Y_{,\mu}, Y_{;\mu\nu}\}, \qquad \mu = 0, \dots, 3, \quad \mu \le \nu, \tag{4}$$

are linearly independent. The ten $Y_{;\mu\nu}$ are normal to the four $Y_{,\mu}$.

Theorem 2.2 (Profile lift of a free embedding)**.** *Let $Y : U \to \mathbb{E}^{13,1}$ be a local free isometric embedding of a Lorentzian metric g. Define one biquaternionic profile field*

$$\boxed{\Theta_Y(x, \psi) := \sqrt{\mathcal{N}_0}\, Y^A(x) F_A(\psi).} \tag{5}$$

Then:

1. *$g_{\mu\nu}[\Theta_Y] = g_{\mu\nu}$ exactly;*
2. *the normal profile second derivatives are*
$$B_{\mu\nu} := P_\perp(\nabla_\mu E_\nu) = \sqrt{\mathcal{N}_0}\, \iota(Y_{;\mu\nu}); \tag{6}$$
3. *the closure map*
$$\mathcal{K}(X) = X^{\mu\nu} B_{\mu\nu} \tag{7}$$
is injective.

Proof. Differentiating Eq. (5) gives $E_\mu = \sqrt{\mathcal{N}_0}\, \iota(Y_{,\mu})$. The isometry of ι yields

$$\mathcal{N}_0^{-1} \langle E_\mu, E_\nu \rangle_\psi = \eta_{AB} Y^A{}_{,\mu} Y^B{}_{,\nu} = g_{\mu\nu}.$$

Because ι is constant and isometric, it commutes with spacetime covariant differentiation and with orthogonal decomposition relative to the tangent span. Hence Eq. (6) holds. Freeness makes the ten normal vectors $Y_{;\mu\nu}$ linearly independent, and the injectivity of ι preserves that independence. Therefore $\mathcal{K}(X) = 0$ implies $X^{\mu\nu} = 0$. □

Theorem 2.3 (Local representability)**.** *Every four-dimensional Lorentzian manifold of the regularity covered by the local free-embedding theorem admits, near each point, a field of the form Eq. (5) that reproduces its metric and is fiber-free.*

Proof. The local free-isometric-embedding theorem gives a free isometric embedding $Y : U \to \mathbb{E}^{13,1}$. Apply the profile-lift theorem. □

Corollary 2.4 (Local Schwarzschild and general Einstein patches)**.** *Every regular coordinate patch of Schwarzschild, Kerr, FLRW, or any other smooth Lorentzian Einstein solution has a local fiber-free Θ_Y representer in the candidate profile metric. This is an existence theorem; it does not supply a preferred or global closed-form representer.*

3 Gauss identity and a direct pure-Θ action

The target profile space has a fixed flat pairing. Its Gauss equation gives

$$R_{\mu\nu\rho\sigma} = \frac{1}{\mathcal{N}_0}\left(\langle B_{\mu\rho}, B_{\nu\sigma}\rangle_\psi - \langle B_{\mu\sigma}, B_{\nu\rho}\rangle_\psi\right). \tag{8}$$

Consequently

$$R[g[\Theta]] = \frac{1}{\mathcal{N}_0}\left(\langle B^\mu{}_\mu, B^\nu{}_\nu\rangle_\psi - \langle B_{\mu\nu}, B^{\mu\nu}\rangle_\psi\right). \tag{9}$$

Definition 3.1 (Pure-Θ Gauss action)**.** *On the regular profile branch define*

$$\boxed{S_{\mathrm{G}}[\Theta] = \frac{1}{2\kappa}\int_U \sqrt{-g[\Theta]}\left[\frac{1}{\mathcal{N}_0}\left(\langle B^\mu{}_\mu, B^\nu{}_\nu\rangle_\psi - \langle B_{\mu\nu}, B^{\mu\nu}\rangle_\psi\right) - 2\Lambda\right]\mathrm{d}^4x.} \tag{10}$$

The usual composite boundary term is understood, or variations are taken with compact support.

Proposition 3.2 (No independent metric or connection)**.** *On every regular profile configuration, Eq.* (10) *equals the Einstein–Hilbert–Λ functional of the composite metric $g[\Theta]$. All objects in Eq.* (10) *are built from one Θ, the fixed profile pairing, and spacetime derivatives.*

Theorem 3.3 (Exact local vacuum GR closure)**.** *Let Θ be a regular fiber-free stationary point of $S_{\mathrm{G}}[\Theta]$. Then*

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 0.} \tag{11}$$

Conversely, every local free profile lift of an Einstein–Λ metric is a stationary point under compactly supported profile variations.

Proof. The metric variation is

$$\delta g_{\mu\nu} = \frac{2}{\mathcal{N}_0}\left\langle E_{(\mu}, \partial_{\nu)}\delta\Theta\right\rangle_\psi.$$

Varying the composite Einstein–Hilbert functional, integrating by parts, and using the contracted Bianchi identity gives the normal Euler–Lagrange equation

$$(G^{\mu\nu} + \Lambda g^{\mu\nu})B_{\mu\nu} = 0. \tag{12}$$

Fiber-freeness makes the ten $B_{\mu\nu}$ independent, so Eq. (12) implies Eq. (11). The converse follows immediately because the Einstein residual vanishes before contraction with $B_{\mu\nu}$. □

4 What is closed and what remains

Item	Status	Scope
Local metric representability	Closed (profile metric)	Follows from local free embedding in $\mathbb{E}^{13,1}$ and the profile lift.
Fiber-free non-vacuity on GR geometries	Closed locally	Every locally free embedded GR patch lifts to one Θ.
Vacuum Einstein–Λ equation	Closed (free branch)	Direct variation of the pure-Θ Gauss action.
Independent tetrad/connection fields	None introduced	g, B, and the Levi–Civita objects are composites of Θ.
Canonical metric choice	Axiom decision	Current AXIOM C explicitly forbids compact-ψ averaging.
Origin of κ and Λ	Open	The Gauss action fixes the form, not their values from the older master equation.
Dynamic profile selection/stability	Open	Must show the UBT dynamics selects and preserves the free, ψ-stable branch.
Holomorphic/Jacobi restriction	Open	The lift is smooth periodic; any stronger analytic restriction must be checked separately.
Global representability	Open	The theorem is local; topology and horizons require patching/global analysis.
Matter and gauge separation	Open	Vacuum needs no separate $\mathcal{F}_\Theta$ term; unified matter requires a projection/separation theorem.

5 Decision

The candidate profile completion is not merely a right inverse for a prescribed tetrad. Once the profile metric and the Gauss action are adopted, the field Θ is the embedding variable, its stationary equation is Eq. (12), and freeness converts that equation to the complete ten Einstein equations. The construction therefore supplies an exact local vacuum GR sector from one Θ.

The remaining obstacle is conceptual and canonical rather than a missing rank calculation: decide whether the classical metric is a pointwise tetrad readout or the translation-invariant readout of the full already-existing complex-time profile. The former has the proved torsion-free concurrent-vector no-go; the latter has the exact local closure proved above.

References

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